EEE303 Tutorial 5a

1. A 3-phase generator turning at 1200 rpm generates a no-load voltage of 9 kV, 60 Hz. If the field current is kept constant, calculate the no-load voltage and frequency if the speed is 1000 rpm and 5 rpm.

2. Referring to Figure T5.1, calculate the exciting current needed to generate a no-load line voltage of
   a. 24.2 kV
   b. 12.1 kV

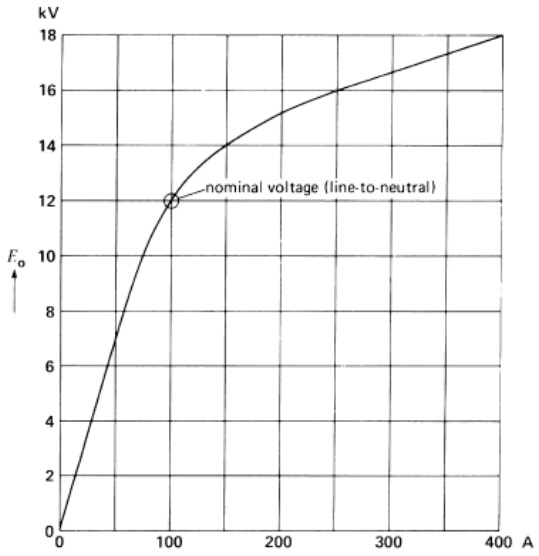


Figure T5.1

3. A 3-phase generator possesses a synchronous reactance of 6 Ω and the excitation voltage E_o is 3 kV per phase Figure T5.2. Calculate the line-to-neutral voltage E and real power for a resistive load of 8 Ω and draw the phasor diagram.

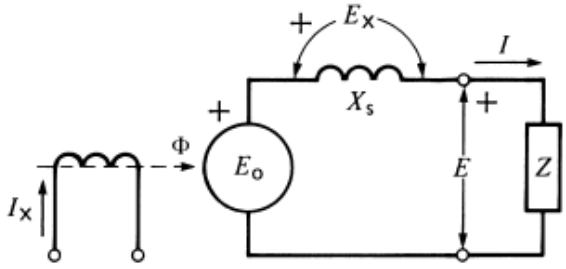


Figure T5.2

4. From Question 3,
   a. Draw the curve of E versus I for the following resistive loads: infinity, 24, 12, 6, 3, 0 Ω.

   b. Calculate the active power P per phase in each case.

   c. Draw the curve of E versus P. For what value of load resistance is the power output a maximum?

5. The 3-phase generator shown in Figure T5.3 has the following characteristics: E_o = 2400V, X_s = 144 Ω, R = 17 Ω, load impedance Z = 175 Ω (resistive). Calculate
   a. The synchronous impedance Zs per phase
   b. The total resistance of the circuit, per phase
   c. The total reactance of the circuit, per phase
   d. The line current
   e. The line-to-neutral voltage across the load
   f. The line voltage across the load
   g. The power of the turbine driving the alternator
   h. The phase angle between Eo and the voltage across the load

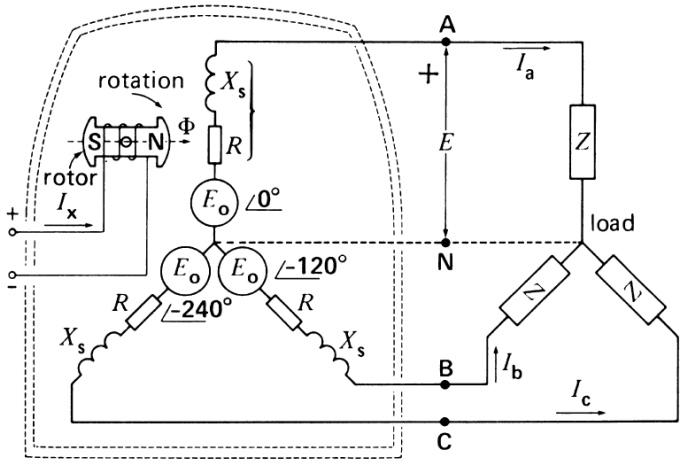


Figure T5.3

6. A 200 kVA, 480 V, 50 Hz, synchronous generator with a rated field current 5 A was tested, and the following data were taken:
   - Open circuit line voltage at rated I_f was measured to be 540 V.
   - Short circuit line current at the rated I_f was found to be 300 A.

   When a dc voltage of 10 V was applied to two of the terminals, a current of 25 A was measured. Calculate the values of the armature resistance and the approximate synchronous reactance, synchronous impedance (all in ohm) and rated current in phase for
   a) Y-connected stator winding
   b) Δ-connected stator winding

   Sketch the per-phase equivalent circuit of the generator and show the parameter value for both armature connection.

7. A 3-phase Y-connected generator rated 3000 KVA (apparent power), 20 kV, 900 r/min, 60 Hz delivers power to a 2400 KVA, 16 kV load having a lagging power factor of 0.8. If the synchronous reactance is 100 Ω, calculate the value of E_o, per phase.